

ASSESSMENT TOOL : 3

CLASS TEST

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1. An image appears too bright. Explain and Justify the most suitable grey level transformation to improve visibility.

A suitable grey level transformation is Negative Transformation. It converts bright pixels into dark pixels and dark into bright using the formula " $s = 1 - r$ ".

WORKING:

Every pixel intensity is replaced by its negative value.

JUSTIFICATION:

This helps reveal hidden details in bright regions and improves image visibility.

EXPECTED OUTCOME:

Better contrast, improved visibility & easier image analysis.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation.

A Log Transformation is used to enhance dark images. It increases low-intensity pixel values more than high-intensity values.

WORKING:

Each pixel is transformed using the logarithmic function,
 $s = c \log(1+r)$.

JUSTIFICATION:

It brightens dark regions while maintaining the brightness of other regions.

EXPECTED OUTCOME:

Hidden details become visible, & the overall image quality improves.

3. An image has poor contrast. Explain a suitable histogram-based technique.

The best technique is Histogram Equalization.

WORKING:

It redistributes pixel intensities across the full grey-level range, making the histogram more uniform.

JUSTIFICATION:

This increases image contrast & improves visibility of details.

EXPECTED OUTCOME:

Clearer image, enhanced contrast & better object recognition.

4. A noisy image contains random intensity variations. Explain the most suitable smoothing filter.

A Mean Filter is commonly used for reducing random noise.

WORKING:

It replaces each pixel with the average value of its neighboring pixels.

JUSTIFICATION:

Averaging smooths sudden intensity changes caused by noise.

EXPECTED OUTCOME:

Reduced noise & a smoother image, although slight blurring may occur.

5. An image requires noise reduction without significant loss of details. Describe an appropriate filtering approach.

The Median Filter is the best choice.

WORKING:

It replaces the center pixel with the median value of neighboring pixels.

JUSTIFICATION:

It removes noise while preserving edges better than the mean filter.

EXPECTED OUTCOME:

Clean image with reduced noise and preserved details.

6. An image appears blurred after smoothing. Explain a suitable sharpening method.

The Laplacian Filter is used for image sharpening.

WORKING:

It detects edges by highlighting rapid intensity changes and enhances them.

JUSTIFICATION:

It restores details lost during smoothing.

EXPECTED OUTCOME:

Sharper edges & improved image clarity.

7. An object is clearly brighter than the background. Explain the suitable segmentation technique.

The suitable technique is Thresholding.

WORKING:

A threshold value is selected. Pixels above the threshold become the object, while others become the background.

JUSTIFICATION:

It is simple & effective when the object is brighter than the background.

EXPECTED OUTCOME:

Accurate separation of the object from the background.

8. You need to separate foreground and background based on intensity. Describe the method.

The appropriate method is Global thresholding.

WORKING:

A single threshold value is applied to the entire image to classify foreground & background.

JUSTIFICATION:

It works well for images with uniform lighting.

EXPECTED OUTCOME:

Easy & accurate image segmentation.

An image requires identification of object edges. Explain a suitable edge detection method.

The Sobel Edge Detection method is commonly used.

WORKING:

It calculates horizontal & vertical intensity gradients to detect edges.

JUSTIFICATION:

It accurately identifies object boundaries while reducing noise.

EXPECTED OUTCOME:

Clear edge detection & improved feature extraction.

10. An image contains salt-and-pepper noise. Explain the most appropriate spatial filter.

The Median Filter is the most suitable filter for removing salt-and-pepper noise.

WORKING:

It replaces noisy pixels within the median value of neighboring pixels.

JUSTIFICATION:

It effectively removes impulse noise without blurring image edges.

EXPECTED OUTCOME:

Noise-free image with preserved edges & important details.